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AGILEX SCOUT MINI AMR SIMULATION

Engineering Troubleshooting, Failure Analysis & Root Cause Case Study
Technical Log of All 22 System Anomalies, Error Signatures, Root Causes & Engineering Solutions

Course & Subject:	M.E. Mechatronics / RA23E01: Robot Operating System (ROS)	**Candidate Name:**	Sailakshmi B.T. (Reg No: 2024511019)
Target Robot Platform:	AgileX Scout Mini 4-Wheel Skid-Steer AMR	**Software Ecosystem:**	Ubuntu 24.04 LTS | ROS 2 Jazzy | Gazebo Harmonic
Document Scope:	Rigorous failure logs, root cause diagnostics, and engineering fixes across CAD, physics, DDS, EKF, and Nav2.	**Format Specification:**	Exhaustive Text-Based Troubleshooting Log (Zero Screenshots)

Executive Engineering Summary: In advanced mobile robotics, transitioning a mechanical SolidWorks CAD assembly into a fully validated autonomous simulation within ROS 2 Jazzy Jalisco and Gazebo Harmonic (`gz-sim8`) presents complex multi-domain challenges. The AgileX Scout Mini project encountered **22 distinct system failures** spanning Linux operating system standards, SolidWorks kinematic joint exports, ODE contact physics micro-collisions, network multicast routing, sensor frame transformations, Extended Kalman Filter fusion, and ROS 2 Jazzy Nav2 lifecycle management. This document provides an exhaustive, text-based engineering post-mortem: every failure is paired with its exact error traceback, in-depth root cause analysis explaining the underlying theoretical mechanism, and the permanent applied solution.

Summary Matrix of All 22 System Anomalies & Engineering Root Causes

#	Domain	Failure Phenomenon	Primary Root Cause / Failure Reason
#1	ROS 2 Build	**ROS 2 Package Naming Convention Failure (ValueError)**	The initial directory and package inherited the SolidWorks assembly notation 'assem2.robot'. In ROS 2, ament a...
#2	OS / File	**Linux Ext4 Case-Sensitivity & File Resolution (FileNotFoundError)**	Windows and SolidWorks are case-insensitive, but Linux Ext4 file systems are strictly case-sensitive. SolidWor...
#3	URDF / Mesh	**Broken Mesh URI Resolution & Invisible Robot in RViz**	The SolidWorks URDF exporter hardcoded the export assembly workspace directory name ('Assem2.SLDASM') into all...
#4	Kinematics	**SolidWorks Fixed Wheel Joints & Missing TF Transforms**	The SolidWorks URDF exporter defaulted unconstrained or mated rotational components to type='fixed' joints ins...
#5	Visualization	**Empty RViz Fixed Frame Dropdown & Blank Display**	RViz launched with a blank default layout lacking the RobotModel display plugin. Furthermore, by default RViz ...
#6	Network / DDS	**DDS Network Multicast Failures & Missing Odometry Packets**	The default FastDDS discovery mechanism attempted to broadcast peer discovery multicast packets across externa...
#7	ODE Physics	**Wheel 3 (j3) Infinite Spinning Physics Bug at Rest**	Three compounding root causes: 1) The SolidWorks export omitted the tag inside joint j3; 2) Zero joint ...
#8	Architecture	**ROS 2 Humble to Jazzy / Gazebo Classic to Harmonic Shift**	Ubuntu 24.04 (Noble) and ROS 2 Jazzy completely retired Gazebo Classic (gazebo11). The simulation engine shift...
#9	Kinematics	**Robot Sunken 23 cm Below RViz Ground Grid**	SolidWorks placed the coordinate origin at the top cover plate of the chassis. Wheel axle height was located a...
#10	KDL Kinematics	**KDL Parser Warning on Root Link Mass and Inertia**	The Kinematics and Dynamics Library (KDL) tree algorithm utilized by robot_state_publisher mathematically requ...
#11	Simulation	**Gazebo Duplicate Spawning ('Two Bots' Collision Bug)**	The ros_gz_sim create node defaults to allow_renaming: true. If a previous simulation run was interrupted or c...
#12	Build System	**Disk Storage Footprint Management on Limited Partition**	Standard colcon build copies entire binary trees and STL meshes into both build/ and install/ directories, dup...
#13	Coordinate REP	**Teleoperation Heading Mismatch (+Y CAD Heading vs +X ROS Heading)**	In SolidWorks, the CAD designer aligned the front bumper with the +Y axis. ROS standard REP-103 strictly requi...
#14	Physics Dynamics	**Wheel Rotation Axis Inversion & Direction Discrepancy**	The CAD export assigned right-hand rule axes to wheel continuous joints that opposed the physical forward rota...
#15	Sensors	**RGB-D Camera Inversion (Mounted on Rear Instead of Front)**	Due to the CAD origin orientation, the physical front bumper was located at negative Y in CAD coordinates (Y =...
#16	Localization	**Severe Skid-Steer Rotational Odometry Drift**	Skid-steering relies on physical tire slipping and scrubbing against the ground. In-hub wheel encoders only me...
#17	Time / Clock	**Dual /clock Publisher Timestamp Oscillations & TF Jitter**	Both ros_gz_sim (gz_sim.launch.py) and the parameter_bridge were publishing to the /clock topic with slight mi...
#18	ROS Lifecycle	**SLAM Toolbox Lifecycle State Unconfigured Error**	In ROS 2 Jazzy, SLAM Toolbox is implemented as a Managed Lifecycle Node. Starting the raw node binary leaves i...
#19	TF Buffer	**RViz 'Frame [odom] does not exist' on Base Simulation Launch**	RViz default Fixed Frame was set to 'map', which only exists when SLAM or AMCL are active....
#20	Localization	**AMCL Initial Pose Frame Mismatch**	The initial pose was published in the 'odom' frame instead of the global 'map' frame....
#21	Nav2 Stack	**ROS 2 Jazzy Nav2 Behavior Tree Navigator Architecture Shift**	ROS 2 Jazzy refactored Nav2's bt_navigator parameters: the older single-navigator configuration was replaced w...
#22	Mission Control	**Autonomous Patrol Mission Feedback & Telemetry Supervision**	BasicNavigator required explicit polling of isTaskComplete() and handling of TaskResult.FAILED to transition s...

Detailed Chronological Failure Cases, Root Cause Diagnostics & Solutions

The following sections record each engineering failure in full technical depth, detailing the exact failure symptoms, underlying root causes, and permanent solutions.

FAILURE CASE #1: ROS 2 PACKAGE NAMING CONVENTION FAILURE (VALUEERROR)	DOMAIN: **ROS 2 Build**
Observed Error Symptom / Diagnostic Traceback:	`ValueError: 'assem2.robot' is not a valid package name. InvalidFrontendLaunchFileError.`
Root Cause Analysis (Failure Reason):	The initial directory and package inherited the SolidWorks assembly notation 'assem2.robot'. In ROS 2, ament and Python packaging rules strictly enforce that package names must contain only lowercase alphanumeric characters and underscores. Periods ('.') and hyphens ('-') are illegal characters that cause parsing errors in ament_python and colcon.
Applied Permanent Engineering Solution:	Standardized the package name across package.xml, CMakeLists.txt, launch scripts, and URDF references to 'assem2_robot'.

FAILURE CASE #2: LINUX EXT4 CASE-SENSITIVITY & FILE RESOLUTION (FILENOTFOUNDERROR)	DOMAIN: **OS / File**
Observed Error Symptom / Diagnostic Traceback:	`FileNotFoundError: [Errno 2] No such file or directory: '.../urdf/assem2.SLDASM.urdf'`
Root Cause Analysis (Failure Reason):	Windows and SolidWorks are case-insensitive, but Linux Ext4 file systems are strictly case-sensitive. SolidWorks exported the file with a capitalized 'A' (Assem2.SLDASM.urdf), while the ROS 2 launch script referenced lowercase 'assem2.SLDASM.urdf'. Additionally, CMakeLists.txt lacked an install(DIRECTORY urdf ...) macro, omitting the files from install/.
Applied Permanent Engineering Solution:	Synchronized launch file paths to exact casing 'Assem2.SLDASM.urdf' and added 'install(DIRECTORY urdf DESTINATION share/${PROJECT_NAME})' in CMakeLists.txt.

FAILURE CASE #3: BROKEN MESH URI RESOLUTION & INVISIBLE ROBOT IN RVIZ	DOMAIN: **URDF / Mesh**
Observed Error Symptom / Diagnostic Traceback:	`Could not load resource [package://Assem2.SLDASM/meshes/base_link.STL]. Robot links were completely invisible in RViz2 and Gazebo.`
Root Cause Analysis (Failure Reason):	The SolidWorks URDF exporter hardcoded the export assembly workspace directory name ('Assem2.SLDASM') into all <mesh filename="package://..."> tags. When the package was standardized to 'assem2_robot', all mesh file lookups failed because the package name in the URI did not exist in the ROS environment.
Applied Permanent Engineering Solution:	Executed a global regex substitution across Assem2.SLDASM.urdf replacing 'package://Assem2.SLDASM/' with 'package://assem2_robot/', restoring 100% mesh visual rendering.

FAILURE CASE #4: SOLIDWORKS FIXED WHEEL JOINTS & MISSING TF TRANSFORMS	DOMAIN: **Kinematics**
Observed Error Symptom / Diagnostic Traceback:	`RViz warning: 'No transform from [w1] to [base_link]'. Robot wheels appeared frozen or detached from the chassis body.`
Root Cause Analysis (Failure Reason):	The SolidWorks URDF exporter defaulted unconstrained or mated rotational components to type='fixed' joints instead of type='continuous'. Since fixed joints have 0 degrees of freedom, joint_state_publisher ignores them and never broadcasts dynamic wheel TF transforms across /tf.
Applied Permanent Engineering Solution:	Modified URDF joints j1, j2, j3, and j4 to type='continuous' with explicit rotation axis <axis xyz='1 0 0'/>, allowing joint_state_publisher to compute and broadcast active wheel transforms.

FAILURE CASE #5: EMPTY RVIZ FIXED FRAME DROPDOWN & BLANK DISPLAY	DOMAIN: **Visualization**
Observed Error Symptom / Diagnostic Traceback:	`Opening RViz showed a blank black viewport. The Fixed Frame dropdown was completely empty, and manually typing base_link reported 'Frame does not exist'.`
Root Cause Analysis (Failure Reason):	RViz launched with a blank default layout lacking the RobotModel display plugin. Furthermore, by default RViz RobotModel searches for a local URDF file on disk instead of subscribing to the live ROS 2 /robot_description topic broadcast by robot_state_publisher.
Applied Permanent Engineering Solution:	Authored a pre-configured display.rviz configuration file with Fixed Frame set to odom/map, RobotModel subscribed to /robot_description, and integrated LaserScan, PointCloud2, and TF visualizers.

FAILURE CASE #6: DDS NETWORK MULTICAST FAILURES & MISSING ODOMETRY PACKETS	DOMAIN: **Network / DDS**
Observed Error Symptom / Diagnostic Traceback:	`Console spam: 'Exception sending a multicast message: Network is unreachable'. Odometry updates dropped intermittently and nodes failed discovery.`
Root Cause Analysis (Failure Reason):	The default FastDDS discovery mechanism attempted to broadcast peer discovery multicast packets across external network interfaces (such as unconfigured Wi-Fi or cellular adapters). When operating offline or on restricted university subnets, multicast packets were dropped.
Applied Permanent Engineering Solution:	Set ROS_AUTOMATIC_DISCOVERY_RANGE=LOCALHOST and exported ROS_LOCALHOST_ONLY=1, restricting all DDS participant discovery and RTPS transport exclusively to the local loopback interface (127.0.0.1).

FAILURE CASE #7: WHEEL 3 (J3) INFINITE SPINNING PHYSICS BUG AT REST	DOMAIN: **ODE Physics**
Observed Error Symptom / Diagnostic Traceback:	`When robot spawned at rest (cmd_vel = 0), wheels 1, 2, and 4 stayed motionless, but Wheel 3 (j3 / w3) spun continuously at maximum velocity, violently oscillating the chassis.`
Root Cause Analysis (Failure Reason):	Three compounding root causes: 1) The SolidWorks export omitted the <axis> tag inside joint j3; 2) Zero joint damping and zero friction existed in the URDF; 3) The initial DiffDrive plugin only controlled joints j1 and j2, leaving j3 completely unpowered and unbraked. ODE micro-collisions with the ground accelerated the frictionless, unconstrained joint into an infinite energy amplification loop.
Applied Permanent Engineering Solution:	Added <axis xyz='-1 0 0'/>, injected <dynamics damping='0.1' friction='0.1'/>, and grouped all 4 wheels into the Gazebo DiffDrive plugin (<left_joint>j2</left_joint><left_joint>j4</left_joint><right_joint>j1</right_joint><right_joint>j3</right_joint>) so active braking torque is applied at zero velocity.

FAILURE CASE #8: ROS 2 HUMBLE TO JAZZY / GAZEBO CLASSIC TO HARMONIC SHIFT	DOMAIN: **Architecture**
Observed Error Symptom / Diagnostic Traceback:	`gazebo_ros plugins failed to compile and launch. Terminal error: 'gazebo: command not found'.`
Root Cause Analysis (Failure Reason):	Ubuntu 24.04 (Noble) and ROS 2 Jazzy completely retired Gazebo Classic (gazebo11). The simulation engine shifted to Gazebo Harmonic (gz-sim8), requiring the new ros_gz_bridge architecture rather than legacy libgazebo_ros_diff_drive.so C++ libraries.
Applied Permanent Engineering Solution:	Replaced legacy C++ .so plugins with native Gazebo Harmonic systems: gz::sim::systems::DiffDrive, gz::sim::systems::Sensors (gpu_lidar), and configured ros_gz_bridge parameter_bridge for topic translation.

FAILURE CASE #9: ROBOT SUNKEN 23 CM BELOW RVIZ GROUND GRID	DOMAIN: **Kinematics**
Observed Error Symptom / Diagnostic Traceback:	`In RViz, the robot appeared half-buried beneath the grid floor (Z < 0), with the ground plane slicing through the middle of the chassis body.`
Root Cause Analysis (Failure Reason):	SolidWorks placed the coordinate origin at the top cover plate of the chassis. Wheel axle height was located at Z = -0.1469m, and tire radius was 0.08m, placing ground contact at Z = -0.2269m. With zero coordinate offset, the top cover sat on the floor plane.
Applied Permanent Engineering Solution:	Introduced root link 'base_footprint' with fixed joint offset xyz='-0.10185 0.16662 0.2269', lifting the chassis so all 4 tires sit flush on the ground at Z = 0.00m.

FAILURE CASE #10: KDL PARSER WARNING ON ROOT LINK MASS AND INERTIA	DOMAIN: **KDL Kinematics**
Observed Error Symptom / Diagnostic Traceback:	`Warning: 'The root link base_link has an inertia specified in the URDF, but KDL does not support a root link with an inertia.'`
Root Cause Analysis (Failure Reason):	The Kinematics and Dynamics Library (KDL) tree algorithm utilized by robot_state_publisher mathematically requires the root link of a kinematic chain to be massless and inertialess. SolidWorks exported base_link as the root while attaching the entire 18 kg chassis mass to it.
Applied Permanent Engineering Solution:	Established 'base_footprint' (a dummy link with 0 mass and no <inertial> tag) as the kinematic root, attaching base_link as its child via a fixed transform.

FAILURE CASE #11: GAZEBO DUPLICATE SPAWNING ('TWO BOTS' COLLISION BUG)	DOMAIN: **Simulation**
Observed Error Symptom / Diagnostic Traceback:	`Launching simulation showed two colliding robot models superimposed on top of each other (assem2_robot and assem2_robot_0), causing explosive physics contacts.`
Root Cause Analysis (Failure Reason):	The ros_gz_sim create node defaults to allow_renaming: true. If a previous simulation run was interrupted or crashed without killing background gz processes, the new launch spawned a duplicate instance and renamed it.
Applied Permanent Engineering Solution:	Added '-allow_renaming false' to the create node arguments, explicit spawning coordinates (Z=0.02m), and configured cleanup scripts (pkill -9 -f 'ros_gz|gz sim') to terminate stale instances.

FAILURE CASE #12: DISK STORAGE FOOTPRINT MANAGEMENT ON LIMITED PARTITION	DOMAIN: **Build System**
Observed Error Symptom / Diagnostic Traceback:	`System warnings: primary Linux partition rapidly running low on free disk space (< 30 GB available).`
Root Cause Analysis (Failure Reason):	Standard colcon build copies entire binary trees and STL meshes into both build/ and install/ directories, duplicating large assets. Additionally, Gazebo simulation logs and core dumps accumulated gigabytes of unmonitored data.
Applied Permanent Engineering Solution:	Used 'colcon build --symlink-install' to create symbolic links instead of duplicating files, cleaned Gazebo log caches, and enforced a strict .gitignore, compressing the repository footprint to ~3.3 MB.

FAILURE CASE #13: TELEOPERATION HEADING MISMATCH (+Y CAD HEADING VS +X ROS HEADING)	DOMAIN: **Coordinate REP**
Observed Error Symptom / Diagnostic Traceback:	`Sending linear velocity (+X) made the robot drive sideways; turning commands rotated about an unexpected lateral axis.`
Root Cause Analysis (Failure Reason):	In SolidWorks, the CAD designer aligned the front bumper with the +Y axis. ROS standard REP-103 strictly requires +X as forward, +Y as left, and +Z as upward.
Applied Permanent Engineering Solution:	Applied a +90° (+1.5708 rad) yaw rotation in base_footprint_joint: origin rpy='0 0 1.5707963', aligning the CAD bumper perfectly with the ROS forward direction.

FAILURE CASE #14: WHEEL ROTATION AXIS INVERSION & DIRECTION DISCREPANCY	DOMAIN: **Physics Dynamics**
Observed Error Symptom / Diagnostic Traceback:	`In Gazebo physics, commanding positive linear velocity (+X) caused the robot to roll backward, while RViz showed forward odometry.`
Root Cause Analysis (Failure Reason):	The CAD export assigned right-hand rule axes to wheel continuous joints that opposed the physical forward rotation direction of the wheels.
Applied Permanent Engineering Solution:	Calibrated wheel joint axes: set <axis xyz='1 0 0'/> for left wheels (j1, j2) and <axis xyz='-1 0 0'/> for right wheels (j3, j4), synchronizing physics with kinematics.

FAILURE CASE #15: RGB-D CAMERA INVERSION (MOUNTED ON REAR INSTEAD OF FRONT)	DOMAIN: **Sensors**
Observed Error Symptom / Diagnostic Traceback:	`RViz camera feed showed the arena behind the robot rather than the forward path.`
Root Cause Analysis (Failure Reason):	Due to the CAD origin orientation, the physical front bumper was located at negative Y in CAD coordinates (Y = -0.355m). An initial positive offset mounted the sensor on the rear bumper.
Applied Permanent Engineering Solution:	Mounted camera_mount_joint at xyz='-0.16662 -0.355 0.065' with rpy='0 0 -1.5708', placing the camera squarely on the physical front bumper facing forward.

FAILURE CASE #16: SEVERE SKID-STEER ROTATIONAL ODOMETRY DRIFT	DOMAIN: **Localization**
Observed Error Symptom / Diagnostic Traceback:	`After executing a 360° turn, raw wheel odometry reported only 240° of rotation (a 120° heading error). In RViz, laser scans smeared and distorted.`
Root Cause Analysis (Failure Reason):	Skid-steering relies on physical tire slipping and scrubbing against the ground. In-hub wheel encoders only measure wheel rotations, completely missing lateral scrubbing and kinematic track width expansion.
Applied Permanent Engineering Solution:	Implemented an Extended Kalman Filter (robot_localization ekf_node) fusing raw wheel odometry (/odom) with the 100Hz rate gyroscope from the 6-axis IMU (/imu), yielding drift-free heading on /odometry/filtered.

FAILURE CASE #17: DUAL /CLOCK PUBLISHER TIMESTAMP OSCILLATIONS & TF JITTER	DOMAIN: **Time / Clock**
Observed Error Symptom / Diagnostic Traceback:	`RViz displays flickered rapidly. TF buffer reported timestamps jumping backward in time and transform lookups failed.`
Root Cause Analysis (Failure Reason):	Both ros_gz_sim (gz_sim.launch.py) and the parameter_bridge were publishing to the /clock topic with slight microsecond phase offsets, creating competing simulation clocks.
Applied Permanent Engineering Solution:	Removed the duplicate clock bridge from parameter_bridge arguments, leaving gz_sim as the single authoritative source of simulation time.

FAILURE CASE #18: SLAM TOOLBOX LIFECYCLE STATE UNCONFIGURED ERROR	DOMAIN: **ROS Lifecycle**
Observed Error Symptom / Diagnostic Traceback:	`SLAM node spawned but failed to publish /map or /tf. Node reported an unconfigured lifecycle state.`
Root Cause Analysis (Failure Reason):	In ROS 2 Jazzy, SLAM Toolbox is implemented as a Managed Lifecycle Node. Starting the raw node binary leaves it in the 'Unconfigured' state without transitioning to 'Active'.
Applied Permanent Engineering Solution:	Included the official 'online_async_launch.py' with 'autostart: true', which uses nav2_lifecycle_manager to automatically transition SLAM to the 'Active' state.

FAILURE CASE #19: RVIZ 'FRAME [ODOM] DOES NOT EXIST' ON BASE SIMULATION LAUNCH	DOMAIN: **TF Buffer**
Observed Error Symptom / Diagnostic Traceback:	`RViz showed red TF transform errors on startup whenever SLAM or Nav2 were not enabled.`
Root Cause Analysis (Failure Reason):	RViz default Fixed Frame was set to 'map', which only exists when SLAM or AMCL are active.
Applied Permanent Engineering Solution:	Configured default Fixed Frame to 'odom' in display.rviz, and ensured the EKF node automatically publishes the odom -> base_footprint transform.

FAILURE CASE #20: AMCL INITIAL POSE FRAME MISMATCH	DOMAIN: **Localization**
Observed Error Symptom / Diagnostic Traceback:	`AMCL failed to initialize particles; laser scans did not align with pre-mapped walls.`
Root Cause Analysis (Failure Reason):	The initial pose was published in the 'odom' frame instead of the global 'map' frame.
Applied Permanent Engineering Solution:	Set global_frame_id: 'map', odom_frame_id: 'odom', and configured set_initial_pose: true with initial coordinates (x=0, y=0, yaw=0) in nav2_params.yaml.

FAILURE CASE #21: ROS 2 JAZZY NAV2 BEHAVIOR TREE NAVIGATOR ARCHITECTURE SHIFT	DOMAIN: **Nav2 Stack**
Observed Error Symptom / Diagnostic Traceback:	`bt_navigator threw schema errors on startup: 'navigators parameter must be a list of plugins'.`
Root Cause Analysis (Failure Reason):	ROS 2 Jazzy refactored Nav2's bt_navigator parameters: the older single-navigator configuration was replaced with a modular plugin dictionary structure.
Applied Permanent Engineering Solution:	Updated nav2_params.yaml to the official Jazzy format: navigators: ['navigate_to_pose', 'navigate_through_poses'] with their respective plugin definitions.

FAILURE CASE #22: AUTONOMOUS PATROL MISSION FEEDBACK & TELEMETRY SUPERVISION	DOMAIN: **Mission Control**
Observed Error Symptom / Diagnostic Traceback:	`Early patrol scripts hung indefinitely if navigation goal failed to compute a path around blocked stations.`
Root Cause Analysis (Failure Reason):	BasicNavigator required explicit polling of isTaskComplete() and handling of TaskResult.FAILED to transition safely to subsequent waypoints.
Applied Permanent Engineering Solution:	Implemented feedback polling loop at 2Hz reading distance_remaining, with automatic recovery fallback on TaskResult.FAILED.